

Visualization of Complexion for Photo-Realistic Facial Expressions

YoungHee Kim, Woo Jin Jeon, Kwanghyun Shim
Contents Research Division, Electronics and Telecommunications Research Institute

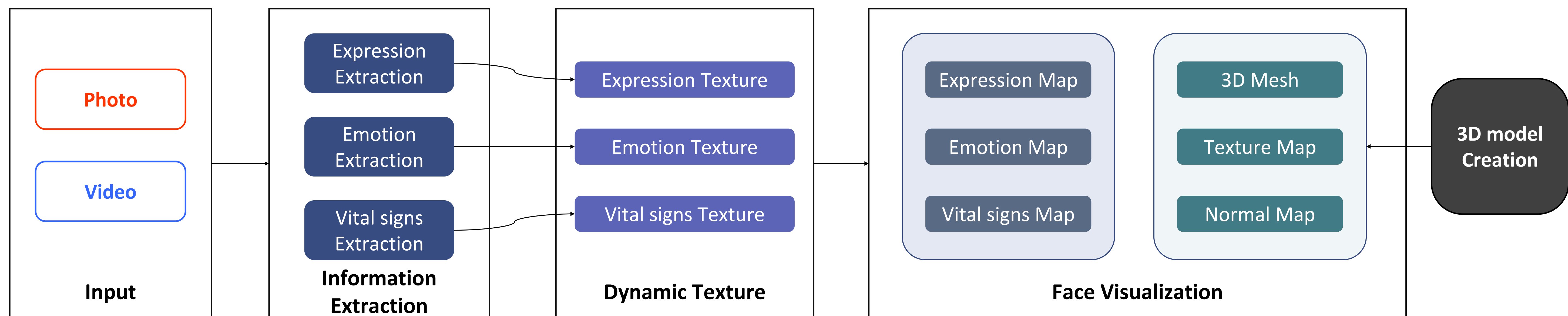


Figure 1: Overall architecture of the face visualization system that extracts facial expressions, emotions, and vital signs from images in real time.

INTRODUCTION

As the metaverse world gradually expands, it is becoming an important concern how realistically we can express digital humans that represent us in the metaverse world. In various industries and research fields, methods for hyper-realistic expression of digital humans are being actively researched. In this paper, we present a novel face visualization system designed to capture and depict dynamic changes in complexion for creating hyper-realistic digital humans. While complexion is often overlooked as a minor detail, it's actually an important part of what makes a digital human recognizable as a person. Facial expressions, emotions, and bio-signals collaboratively influence facial blood flow, resulting in noticeable variations in facial color. By integrating real-time reflection of these intricate components and visualizing them on the digital human's face, we successfully infuse vitality and realism into the digital human representation.

VISUALIZATION OF COMPLEXION REFLECTING BLOOD FLOW

When it comes to facial expression animation, relying solely on geometric movement and fixed face textures has its limitations in achieving hyper-realistic faces of digital humans. When we encounter a digital human or virtual character with an unchanging complexion, we perceive it as unnatural and fail to recognize it as equivalent to a real person (see Fig. 2 (a)). Applying complexion, which continuously changes by various factors, to face visualization plays a crucial role in recognizing digital humans as human beings (see Fig. 2 (b)). To address this, extensive research has been conducted to analyze how blood flow changes during various facial expressions performed by actors and integrate color change maps into facial visualization. In this paper, we introduce a face visualization system that can extract facial expressions, emotions, and vital signs from images in real-time and model blood flow changes to reflect them in face visualization (see Fig. 1).

Visualization of complexion reflecting bio-signals

The dermal layer of the skin contains capillaries, while arterioles are present in the subcutaneous tissue. The changes in facial color resulting from facial expressions described earlier are caused by the movement of blood in the capillaries within the dermis layer of the skin. On the other hand, changes in the facial color associated with bio-signals, such as blood pressure and heart rate, are mainly caused by the flow of blood in the arterioles located in the subcutaneous tissue. The flow of blood due to facial expressions is a localized change around the facial muscles, whereas bio-signals tend to affect the entire face due to alterations in blood circulation. Since light will pass through differently depending on the amount of blood, these changes in blood flow can lead to modifications in light transmission and will change the complexion. Additionally, the thickness of the skin should be taken into accounts. The epidermis and dermis layers, constituting the skin, exhibit non-uniform thickness and can vary 2 to 3 times across different areas of the face. Even when the amount of blood remains constant, areas with thinner skin, such as the cheeks, undergo substantial changes in skin color, whereas regions with thicker skin, like the forehead and the bridge of the nose, experience minimal color variations. In this study, considering the distribution of blood vessels and variations in skin thickness, we model the changes in skin color corresponding to the heartbeat and generate a vital signs map (see Fig. 1 and Fig. 2).

Visualization of complexion reflecting emotions

Emotions have an impact on the physical state of the human body(see Fig. 3). Lauri Nummenmaa showed that the physiological changes associated with different emotions remained consistent across Asian and Western populations, indicating a biological foundation for emotions and their corresponding physical response patterns. Most emotions are associated with increased activity in the upper regions of the body. Fig. 3 shows that the face region shows the most changes. In consequence, this study focuses on modeling facial complexion in response to emotional changes and generating an emotion map (see Fig. 1 and Fig. 2).

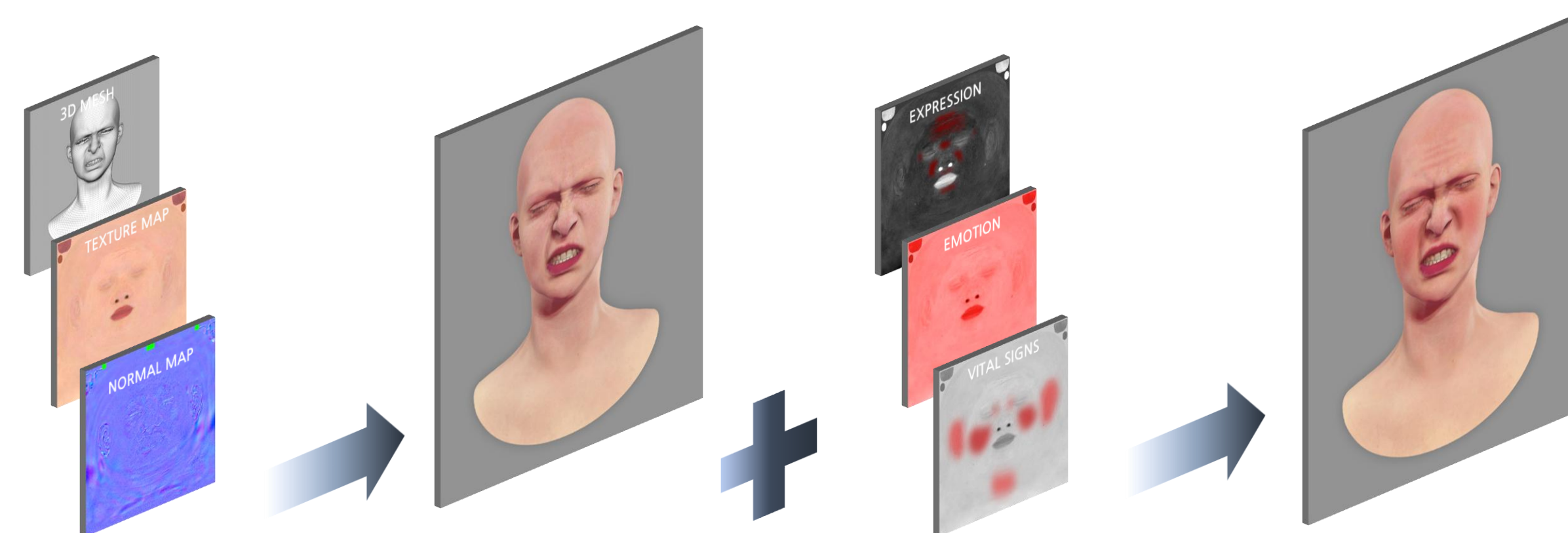


Figure 2: Face visualization with complexion changes. (a) Face visualization without dynamic texture. (b) Face visualization with dynamic texture.

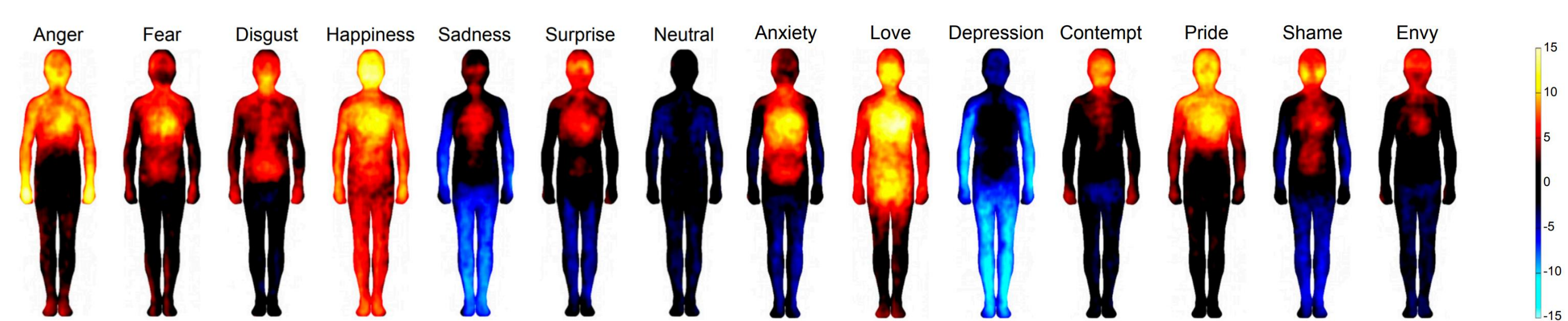


Figure 3: A topographic map of the human body showing the degree of activation of the body according to emotions. Red is the active site and blue is the inactive site. Black means no reaction [1].

REAL-TIME FACIAL COMPLEXION VISUALIZATION SYSTEM

Fig. 1 shows the real-time face visualization system proposed in this work. The system extracts facial expressions, emotions, and vital signs from photos or videos in real-time (see Information Extraction in Fig. 1). Extensive research has been conducted on the extraction of emotions and vital signs from images and has been demonstrating good results. These extracted signals are transformed into textures using respective texture generation models (see Dynamic Texture in Fig. 1). The textures are then converted into Expression maps, Emotion maps, and Vital signs maps (see Face Visualization in Fig. 1). These maps are utilized to reflect real-time changes in complexion of 3D face model, which can be obtained through a 3D scanner or generated using a 3D application (see Face Visualization in Fig. 1 and Fig. 2). Consequently, this system enables the visualization of digital humans with realistic and dynamic complexion.

CONCLUSION

This research presents a face visualization system that aims to achieve hyper-realistic digital humans by accurately portraying dynamic skin color changes influenced by facial blood flow. Despite its often-overlooked nature, complexion plays a crucial role in the recognition of digital humans as human beings. The interplay between facial expressions, emotions, and bio signals contributes to alterations in facial blood flow and subsequently affects the facial color. By incorporating these intricate factors and visualizing them on the digital human face, we successfully enhance the liveliness and realism of the digital representation

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